

Scenario 4E: Strong spatial/covariate confounding stress test with continuous-time x2

MSSTNB posterior performance, confounding calibration, beta recovery, and fit-status
classification

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1 Purpose and main conclusion

Scenario 4E is a strong spatial/covariate confounding stress test for the MSSTNB model under the same continuous-time covariate framework used in the main dynamic simulations. It preserves the reference mean, exposure, negative-binomial dispersion, and temporal latent-risk structure, but regenerates the continuous-time x_2 covariate so that its spatial pattern is strongly aligned with the spatial random effect ϕ .

The main result is:

Under the continuous-time x_2 design, Scenario 4E creates strong x_2 - ϕ confounding while preserving high count information and numerical stability. Across all ten replicates, the fitted model remains stable, the regression slopes remain accurately recovered, and the latent temporal-risk and spatial-effect summaries remain well behaved.

The formal fixed-gamma sampled- λ analysis produced 10 stable fits, 0 soft-warning fits, and 0 numerical-instability fits out of 10 replicates. There were no beta, kappa, lambda-input, or lambda-output numerical guards.

2 Scenario design

Scenario 4E uses the following covariate-confounding construction. The reference dynamic data-generating structure is retained, but x_2 is regenerated as a continuous-time covariate whose spatial mean is aligned with the standardized spatial effect:

$$x_{2,tj}^{\text{conf}} = z_{tj}^{\text{AR}} + c \phi_j^*,$$

where z_{tj}^{AR} is a residual continuous-time AR(1) component, ϕ_j^* is the standardized spatial effect, and c is the confounding strength. The covariate is then centered and scaled so that the resulting x_2 remains continuous with standard deviation one. The intercept is shifted to preserve the reference mean-count scale as closely as possible.

Table 1: Official Scenario 4E continuous-time x2 design.

Quantity	Value
Data scenario ID	S4E_STRONG_..._CONTINUOUS_X2_T100
Fit analysis	Fixed-gamma sampled-lambda fit
Stress source	Strong spatial/covariate confounding
Confounding mechanism	x2 spatial mean aligned with phi
Confounding strength	0.75
Residual AR component	Continuous-time AR(1) residual component
Reference dispersion	r = 15
Stress dispersion	r = 15
Exposure	Same as reference dynamic scenario
Mean structure	Reference beta/phi/lambda structure with beta0 shift preserving mean count
Time points	100

Quantity	Value
Coarse areas	9
x2 mode	continuous_time
x2 AR coefficient	0.50
x2 innovation SD	0.80
Fixed gamma	0.80

The full data scenario ID is S4E_STRONG_SPATIAL_COVARIATE_CONFOUNDING_CONTINUOUS_X2_T100.

This design differs from the other Scenario 4 stress tests. S4A creates a sparse-count regime, S4B creates low and heterogeneous exposure, S4C reduces the negative-binomial dispersion parameter, and S4D shortens the time series. S4E instead keeps the count scale and time length close to the reference dynamic setting but makes the covariate and spatial random effect difficult to distinguish.

3 Confounding and data calibration

The calibration criterion was to create strong x_2 - ϕ confounding without changing the scenario into a sparse-count, low-exposure, overdispersion, or binary-covariate problem. The selected value was $c = 0.75$, the smallest tested value that passed the S4E calibration checks.

Table 2: S4E continuous-time x2 confounding and data-calibration summary.

Quantity	Value
Number of replicates	10
Confounding strength	0.75
Average count	203.972
Reference average count	204.071
Average zero proportion	0.001
Total count	1,835,746
Maximum observed count	2,749
Abs. cell-level cor(x2, phi)	0.575
Reference abs. cell-level cor(x2, phi)	0.055
Increase in abs. cell-level correlation	0.520
Abs. area-mean cor(x2, phi)	0.976
Reference abs. area-mean cor(x2, phi)	0.306
Increase in abs. area-mean correlation	0.669
x2 empirical AR(1)	0.478
x2 SD	1.000
Binary-like x2 proportion	0.000
Average beta0 shift	0.119

The average cell-level absolute correlation between x_2 and ϕ is 0.575, compared with 0.055 in the matched reference design. The area-mean absolute correlation is 0.976, compared with reference value 0.306. Thus, S4E substantially increases spatial/covariate confounding while keeping the average count close to the reference count and the zero proportion near zero.

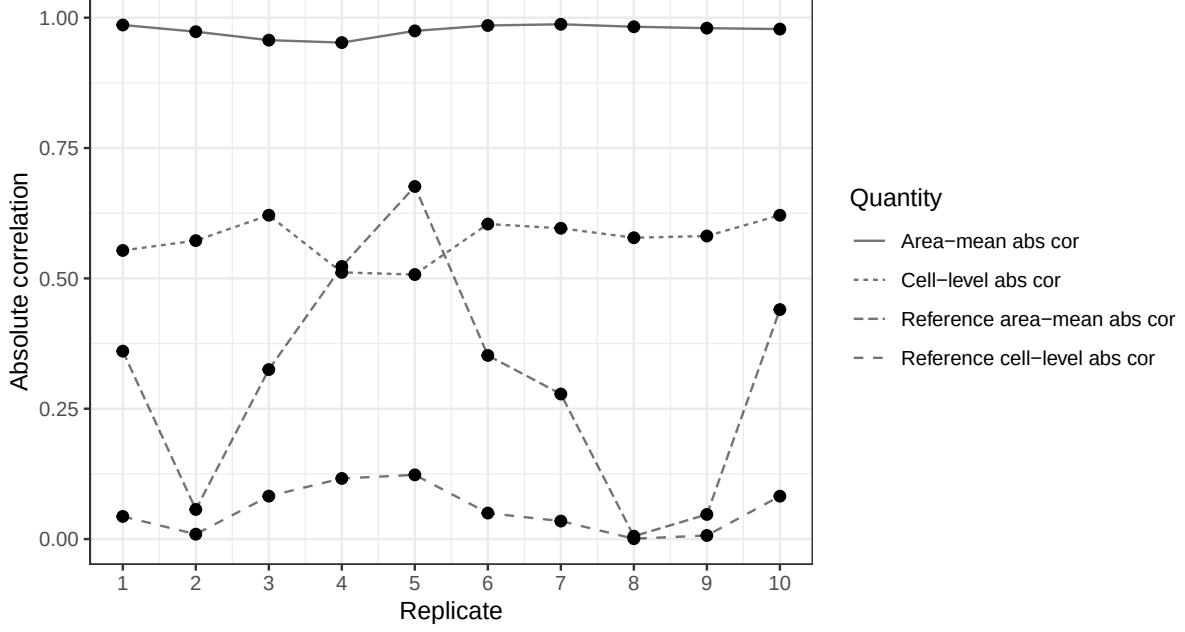


Figure 1: Cell-level and area-mean x2-phi confounding correlations by replicate.

4 Fitting strategy

The S4E fit uses the dynamic MSSTNB structure with fixed $\gamma = 0.8$. The model estimates

$$\beta_0, \beta_1, \beta_2, \phi, \tau_\phi, r, \kappa, \lambda,$$

while γ is fixed at its truth. The split component ω is disabled in this analysis. This fixed-gamma strategy isolates spatial/covariate confounding rather than re-testing weak identification of γ .

Numerical guards are recorded and used to classify each replicate. The classification rule is:

- Stable: no beta, kappa, or lambda guard and no extreme latent-risk metric.
- Soft warning: small guard activity without severe parameter or latent-process failure.
- Numerical instability: large guard activity, extreme coefficient behavior, extreme lambda RMSE, or extreme log-lambda RMSE.

5 Fit-status classification

Table 3: Fit-status classification for S4E continuous-time x2.

Status	# reps	Percent
Stable	10	100.0%

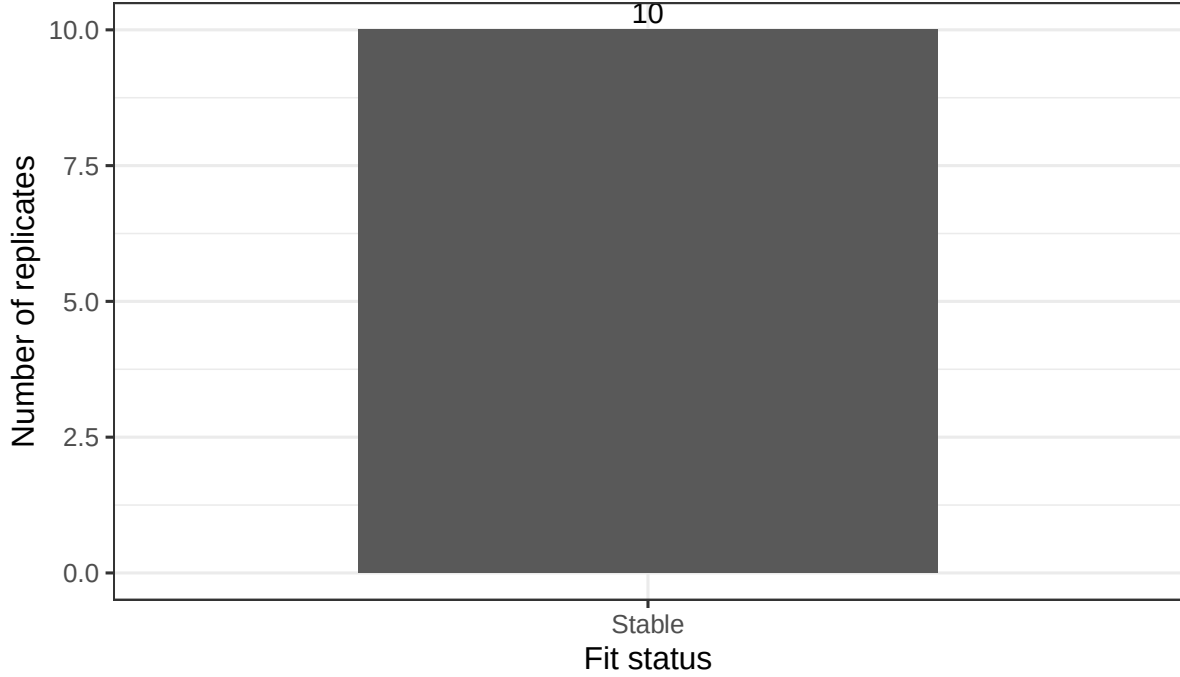


Figure 2: Fit-status counts for S4E continuous-time x2.

All ten formal fits are stable. Therefore, S4E is an identifiability stress test rather than a numerical-failure scenario under the present continuous-time high-count design.

6 Posterior performance

Because all replicates are stable, the all-sampled, stable-only, and stable-plus-soft-warning subsets are identical. The tables below report the main stable set directly.

Table 4: Regression and dispersion recovery for S4E continuous-time x2.

Parameter	Truth	Posterior mean	Across-rep SD	Coverage
beta0	-1.565	-1.566	0.217	1.000
beta1	0.500	0.498	0.012	0.900
beta2	-0.400	-0.398	0.009	1.000
r	15.000	15.286	0.538	0.978

The posterior means for the two slopes are $\bar{\beta}_1 = 0.498$ and $\bar{\beta}_2 = -0.398$, compared with truths $\beta_1 = 0.5$ and $\beta_2 = -0.4$. The empirical coverage is 0.900 for β_1 and 1.000 for β_2 . Thus, in this high-count fixed-gamma setting, strong spatial/covariate confounding does not lead to visible slope-recovery failure.

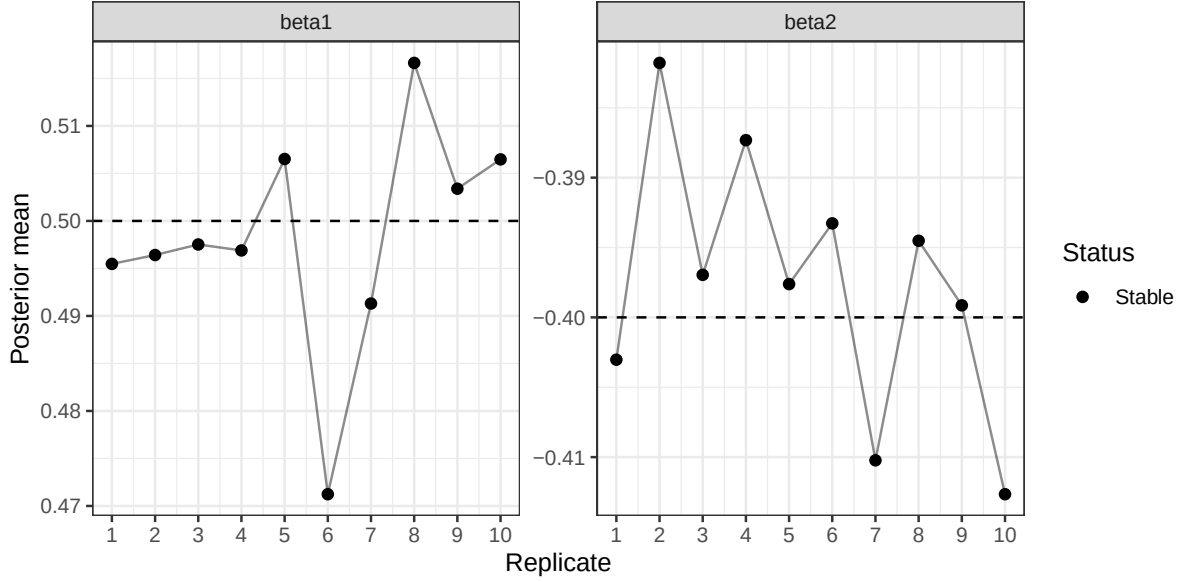


Figure 3: Posterior means of beta1 and beta2 by replicate. Dashed horizontal lines show true values.

7 Replicate-level diagnostics

Table 5: Replicate-level count, confounding, regression, and dispersion summaries.

Rep	Status	Mean count	Zero prop	Abs cell cor	Abs area cor	b0	b1	b2	r
1	Stable	151.833	0.006	0.554	0.986	-1.998	0.495	-0.403	14.799
2	Stable	226.990	0.000	0.572	0.973	-1.332	0.496	-0.382	16.030
3	Stable	210.186	0.000	0.621	0.957	-1.440	0.498	-0.397	15.643
4	Stable	135.289	0.003	0.511	0.952	-1.890	0.497	-0.387	15.185
5	Stable	300.074	0.000	0.507	0.975	-1.381	0.507	-0.398	15.454
6	Stable	211.352	0.000	0.604	0.985	-1.528	0.471	-0.393	14.731
7	Stable	209.347	0.000	0.596	0.987	-1.452	0.491	-0.410	16.122
8	Stable	186.411	0.000	0.578	0.983	-1.636	0.517	-0.395	14.693
9	Stable	194.322	0.000	0.581	0.980	-1.502	0.503	-0.399	14.772
10	Stable	213.913	0.000	0.621	0.978	-1.500	0.506	-0.413	15.432

Table 6: Replicate-level latent-risk and spatial-effect recovery summaries.

Rep	log lambda RMSE	cor(log lambda)	phi RMSE	phi cor
1	0.092	0.911	0.021	1.000
2	0.043	0.797	0.017	0.999
3	0.044	0.759	0.025	0.997
4	0.078	0.924	0.032	1.000
5	0.045	0.615	0.030	0.998
6	0.056	0.820	0.023	1.000

Rep	log lambda RMSE	cor(log lambda)	phi RMSE	phi cor
7	0.056	0.881	0.035	0.999
8	0.058	0.941	0.033	1.000
9	0.042	0.844	0.034	0.996
10	0.049	0.902	0.028	0.999

Table 7: Replicate-level kappa recovery summaries.

Rep	kappa truth CV	kappa post CV	log kappa RMSE	cor(log kappa)
1	0.259	0.228	0.134	0.861
2	0.256	0.236	0.091	0.936
3	0.256	0.235	0.095	0.932
4	0.259	0.226	0.126	0.886
5	0.253	0.239	0.091	0.936
6	0.262	0.245	0.104	0.922
7	0.253	0.227	0.115	0.893
8	0.262	0.237	0.115	0.900
9	0.262	0.242	0.102	0.927
10	0.254	0.236	0.100	0.926

Replicate-level behavior is stable. The confounding correlations are strong in every replicate, but the posterior means for β_1 and β_2 remain close to their truth values. No replicate has beta, kappa, lambda-input, or lambda-output guard activity.

8 Beta-confounding diagnostics

S4E is designed to test whether stronger x_2 - ϕ confounding translates into slope bias or larger slope instability. Because the formal design fixes the confounding strength at 0.75, the range of confounding across the ten replicates is limited. The diagnostic correlations below should therefore be interpreted descriptively rather than as strong inferential evidence.

Table 8: Descriptive regression-bias summaries for S4E.

Parameter	Mean bias	Mean abs. bias
beta0	-0.001	0.005
beta1	-0.002	0.008
beta2	0.002	0.008

Table 9: Descriptive correlations between coefficient bias and replicate-level confounding.

Parameter	Bias-cell cor	Abs-bias-cell cor	Bias-area cor	Abs-bias-area cor
beta0	-0.005	0.066	0.375	0.150
beta1	-0.210	0.263	-0.116	0.483

Parameter	Bias-cell cor	Abs-bias-cell cor	Bias-area cor	Abs-bias-area cor
beta2	-0.403	0.051	-0.466	-0.179

The mean bias is small for all three regression coefficients. In particular, β_2 has mean bias 0.002 and mean absolute bias 0.008. Thus, the main S4E conclusion should not be that confounding produced slope failure. Instead, the result shows that under abundant counts, fixed γ , and the continuous-time x_2 construction, the model can tolerate a substantial increase in spatial/covariate confounding.

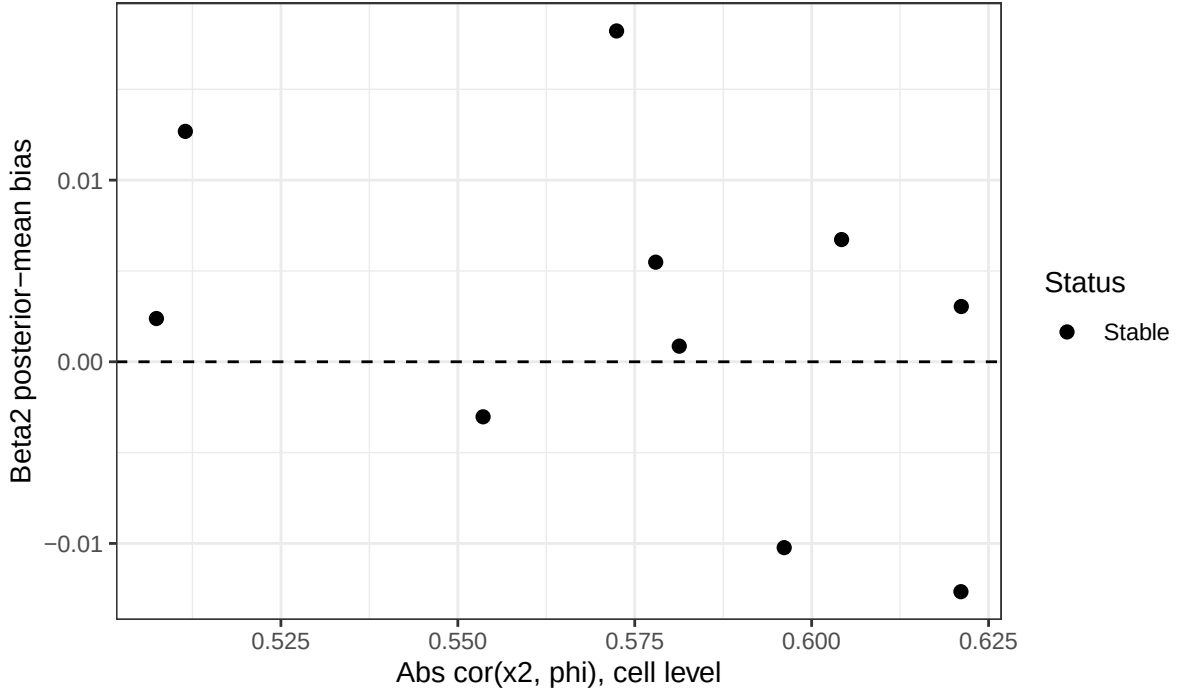


Figure 4: Beta2 bias versus replicate-level x_2 - ϕ cell-level confounding.

9 Dispersion, kappa, and latent-risk recovery

Table 10: Dispersion-parameter recovery for S4E continuous-time x_2 .

Group	N	r truth	r mean	r mean SD	mean bias	mean abs bias	region coverage
All stable reps	10	15.000	15.286	0.538	0.286	0.487	0.978

The average posterior mean of r is 15.286, compared with truth $r = 15$. This is close to the reference dispersion value, indicating that S4E is not an overdispersion stress test.

Table 11: Kappa recovery for S4E continuous-time x2.

Group	N	truth CV	post mean CV	CV bias	kappa RMSE	log kappa RMSE	cor log kappa
All stable reps	10	0.257	0.235	-0.022	0.105	0.107	0.912

Kappa recovery is also stable. The posterior-mean kappa CV is 0.235, compared with truth CV 0.257, and the average log-kappa correlation is 0.912.

Table 12: Latent temporal-risk and spatial-effect recovery for S4E continuous-time x2.

Group	N	lambda RMSE	log lambda RMSE	median log RMSE	cor log lambda	phi RMSE	phi cor
All stable reps	10	0.057	0.056	0.052	0.839	0.028	0.999

The mean log-lambda RMSE is 0.056, and the mean correlation for log lambda is 0.839. Spatial-effect recovery is very strong, with mean ϕ RMSE 0.028 and mean ϕ correlation 0.999.

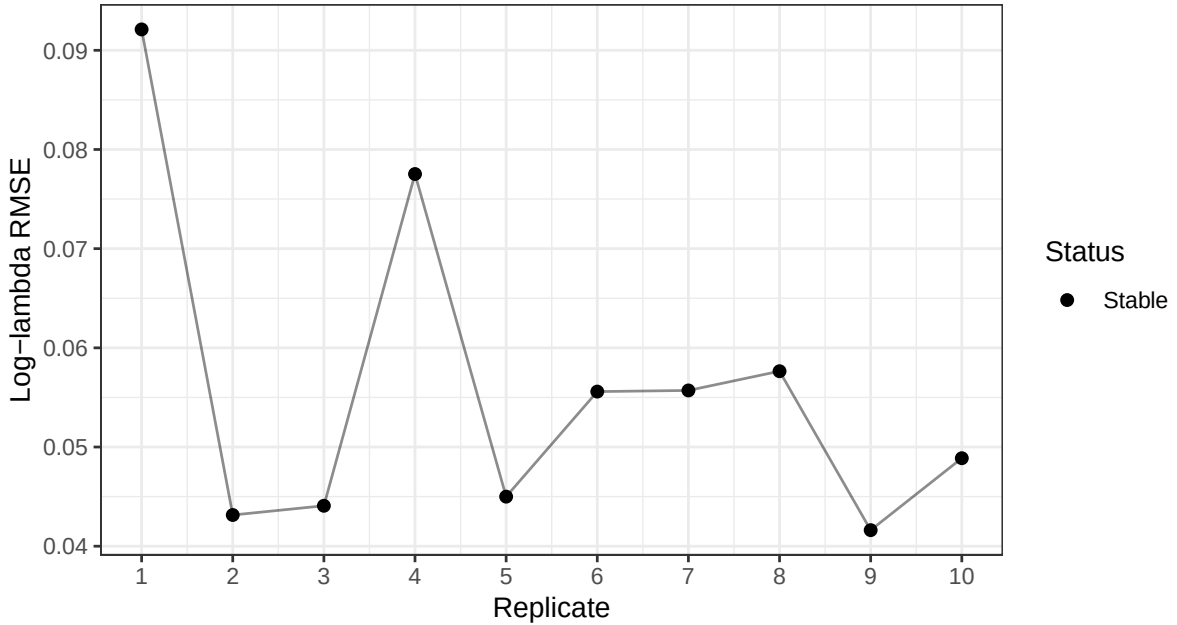


Figure 5: Global log-lambda RMSE by replicate.

10 Numerical guards

Table 13: Numerical guard summary for S4E continuous-time x2.

Group	N	Beta guards	Kappa guards	Lambda input guards	Lambda output guards
All stable reps	10	0	0	0	0

There were no beta guards, no kappa guards, no lambda-input guards, and no lambda-output guards. This confirms that S4E is not a numerical failure scenario under the selected confounding strength.

11 Comparison with the S3 control

Table 14: Comparison of S3 control and S4E: data and confounding metrics.

Scenario	Group	N	Mean count	Zero prop	Abs cell cor	Abs area cor
S3 control	control reps	3	195.050	0.000	–	–
S4E spatial/covariate confounding	stable+soft	10	203.972	0.001	0.575	0.976

Table 15: Comparison of S3 control and S4E: posterior performance metrics.

Scenario	b1 mean	b2 mean	b2 SD	r mean	log RMSE	cor(log)
S3 control	0.492	-0.408	0.009	–	0.047	0.734
S4E spatial/covariate confounding	0.498	-0.398	0.009	15.286	0.056	0.839

Relative to the S3 control, S4E substantially increases x_2 - ϕ confounding while preserving the high-count regime and the continuous-time x_2 structure. The main performance difference is therefore attributable to spatial/covariate confounding rather than a change in sparsity, exposure, dispersion, or time-series length.

12 Interpretation of the stress mechanism

The continuous-time S4E results separate three phenomena.

First, the scenario successfully increases spatial/covariate confounding. The average cell-level absolute correlation between x_2 and ϕ rises from 0.055 in the reference setting to 0.575 in S4E. The area-mean absolute correlation rises from 0.306 to 0.976.

Second, the confounding stress does not create a sparse-count problem. The average count is 203.972 and the average zero proportion is only 0.001.

Third, under this continuous-time high-count fixed-gamma design, slope recovery remains accurate. The posterior means of β_1 and β_2 are close to their true values, and the mean absolute biases are small. Therefore, S4E should be interpreted as a successful spatial/covariate confounding stress with successful recovery, not as a regression-slope failure.

13 Practical implications

The S4E continuous-time x_2 results suggest several practical implications for MSSTNB applications.

1. Spatial/covariate confounding should be calibrated and reported explicitly using both cell-level and area-mean correlations.
2. Strong area-level alignment between a covariate and spatial effect does not necessarily imply coefficient failure when counts are abundant and the covariate retains temporal residual variation.
3. Regression recovery should be interpreted jointly with ϕ recovery, λ recovery, and numerical guards.
4. The selected confounding strength $c = 0.75$ is strong but not degenerate; it avoids making x_2 an almost exact copy of ϕ at the cell level.
5. More severe confounding strengths may still be useful for a boundary analysis, but they would answer a different question from the current S4E design.

14 Final conclusion

Scenario 4E with continuous-time x_2 confirms that the MSSTNB model can remain stable under strong spatial/covariate confounding when count information remains abundant and γ is fixed. The scenario increases average absolute cell-level x_2 - ϕ correlation to 0.575 and average absolute area-mean correlation to 0.976, compared with reference values 0.055 and 0.306.

All ten fixed-gamma sampled- λ fits are stable and no numerical guards occur. The regression slopes are accurately recovered: the posterior means are 0.498 for β_1 and -0.398 for β_2 , compared with truths 0.5 and -0.4. Latent temporal-risk recovery and spatial-effect recovery also remain strong, with mean log-lambda RMSE 0.056 and mean ϕ correlation 0.999.

Therefore, S4E should be interpreted as a strong spatial/covariate confounding stress result with successful recovery under abundant count information, not as a numerical-failure or regression-slope failure scenario.

15 Recommended reporting language

Scenario 4E evaluated the MSSTNB model under strong spatial/covariate confounding while using the same continuous-time x_2 covariate framework as the main dynamic simulations. The continuous-time x_2 covariate was regenerated so that its spatial mean was strongly aligned with the spatial random effect ϕ , producing an average absolute cell-level x_2 - ϕ correlation of 0.575 and an average absolute area-mean correlation of 0.976. The count scale remained close to the reference setting and the zero proportion remained near zero. In the fixed-gamma sampled- λ fit, all ten replicates were stable and no numerical guards occurred. Regression slopes, the dispersion parameter, latent temporal risk, and spatial effects were recovered well. These results indicate that the MSSTNB model is robust to substantial spatial/covariate confounding in this high-count continuous-time setting, although confounding diagnostics should be reported explicitly in real applications.